

INDEPENDENT DATA GOVERNANCE CONSULTANT REPORT

QuickLoan Mobile

Ethical Data Governance Review

Client	Reviewer Role	Scope	Reference Law
QuickLoan Mobile	Data Governance Consultant	Governance, Quality & Ethics	Ghana DPA 2012 (Act 843)

Deliverable 1 — Governance Review Card

Three distinct risk findings from the QuickLoan data pipeline review, each with impact analysis and a concrete mitigation step, plus one reporting metric recommendation for ethical governance.

Section 1 — Data Quality Risk

Issue / Definition	Business Impact	Suggested Fix
Incomplete & Inconsistent Records Loan applicant data (income, phone number, employment) arrives in the Raw Data DB with missing values and mixed formats — e.g., "023-456-7890" vs "0234567890" or "GHS 2000" vs "2000". The Preprocessing Service (Step 5) has no defined handling rules for these cases.	The ML model receives dirty input, producing unreliable credit scores. Creditworthy applicants may be wrongly rejected; risky applicants may be approved. Aggregated reports are inaccurate, undermining business intelligence. Loan officers cannot rely on applicant records for manual escalations.	1. Add input validation at the API Gateway: enforce 10-digit numeric phone format; reject non-numeric income values. 2. Flag records with >20% missing fields for manual review before ML pipeline entry. 3. Implement format standardisation scripts in the Preprocessing Service before model input.

Section 2 — Legal & Compliance Risk

Data Classification

SENSITIVE	QuickLoan collects financial records (mobile money transaction history, income declarations), contact lists, and GPS location — all of which qualify as Sensitive PII under any recognised data classification framework. Sensitive data requires the highest tier of access controls, encryption, and handling restrictions. Its collection demands explicit lawful basis and is subject to strict retention limits under Act 843.
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Compliance Risk Analysis

Compliance Issue	Legal Exposure	Remediation Steps
No Consent Capture & Excessive Data Collection The app collects full contact lists, continuous GPS, and raw device logs with no credit-scoring purpose.	Direct violations of Ghana DPA Act 843: s.14 — Lawful basis & consent required s.17 — Data minimisation principle	A: Data Minimisation — remove contact list, GPS logs. Retain only phone number, income proof, mobile money history.

No explicit informed consent is obtained between collection (Step 1) and storage (Step 3). The Raw Data DB has no classification or retention policy.	s.31 — Retention limits must be defined Penalties: fines and potential criminal liability for data controllers.	B: Consent Layer — add granular consent screen at API Gateway (Step 2) with timestamped consent records. C: Retention Policy — classify Raw DB as Sensitive PII; enforce 24-month automated purge; appoint DPO.
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Section 3 — Bias & Fairness Risk

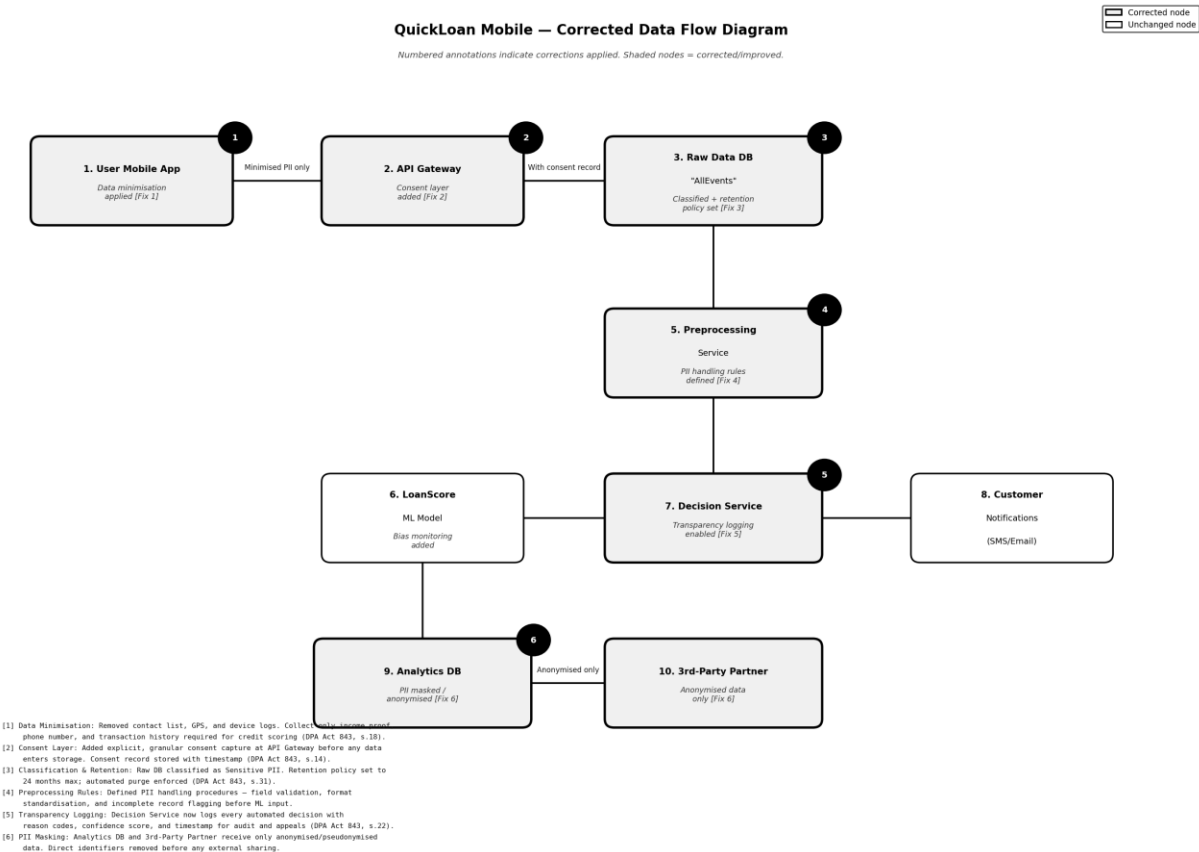
Bias Dimension	Issue & Impact	Monitoring & Mitigation
Source of Bias	Historical Training Data Encoding Systemic Inequalities The LoanScore ML model is trained on historical approval/rejection data that reflects past discriminatory lending patterns. Features such as neighbourhood and device brand act as proxy variables for ethnicity, income class, or gender, producing structurally biased credit scores without any legitimate credit-based justification.	Step 1 — Bias Audit: Run a pre-deployment fairness audit with disaggregated approval rates by gender, age group, and region. Step 2 — Remove Proxy Features: Eliminate neighbourhood and device variables from the feature set.
Operational Impact	Systematically lower approval rates for women, rural applicants, and lower-income areas without legitimate credit-based basis. No transparency logging means customers have no means of understanding or challenging the automated decision — violating Act 843, Section 22 (right to explanation for automated decisions).	Step 3 — Human Review Trigger: For scores within $\pm 10\%$ of the approval threshold, route to a human loan officer before final decision. Step 4 — Ongoing LARDR Monitoring: Track the Loan Approval Rate Disparity Ratio monthly (see Section 4).

Section 4 — Storytelling / Reporting Metric

Metric Name & Definition	Visualization Type	Why It Matters
Loan Approval Rate Disparity Ratio (LARDR) Definition: The ratio of the loan approval rate of the lowest-performing demographic group to the highest-performing group, segmented by gender and region. Formula: $LARDR = \frac{\text{Min(Approval Rate by Group)}}{\text{Max(Approval Rate by Group)}}$ Threshold: Score < 0.80 triggers a mandatory fairness review (four-fifths rule). Score = 1.0 indicates full parity.	Grouped Bar Chart — Monthly X-axis: Demographic segment (gender $\times$ region) Y-axis: Loan approval rate (%) Series: Each monthly reporting period as a grouped bar cluster A horizontal reference line at the minimum acceptable rate (e.g., 70%) makes disparities immediately visible to non-technical executives and regulators without interpretation.	This metric operationalises ethical fairness in a single, auditable number. It makes invisible algorithmic bias visible — translating a complex ML problem into a clear monthly governance signal that board members, regulators, and auditors can understand and act on without needing to understand the ML model's internals. Without this metric, discrimination can persist undetected through every loan cycle indefinitely.

Deliverable 2 — Corrected Data Flow Diagram

The diagram below presents the corrected QuickLoan data pipeline. Shaded nodes indicate components where corrections have been applied. Numbered badges correspond to the corrections table beneath.



Correction Annotations

Each row below corresponds to a numbered badge on the diagram above.

	Pipeline Step	Correction Applied	Rationale (Act 843 Reference)
1	Step 1 — User Mobile App	Data Minimisation: Removed collection of full contact list, continuous GPS, and raw device logs. Retained only phone number, income declaration, and mobile money transaction history.	Violates the Data Minimisation principle (s.17). Excessive collection creates unnecessary legal liability and breach exposure. Only data strictly necessary for credit scoring may be collected.
2	Step 2 → 3 — API Gateway to Raw DB	Consent Capture Layer: Added mandatory granular consent screen at the API Gateway before any personal data enters storage. Consent record (timestamp, version, scope) stored alongside data.	No consent = no lawful basis for processing (s.14). Every downstream operation — scoring, storage, sharing — is unlawful without documented, freely given, specific consent.
3	Step 3 — Raw Data DB "AllEvents"	Data Classification & Retention Policy: Classified DB as Sensitive PII. Implemented 24-month automated	Unclassified DBs cannot have appropriate security controls applied. Act 843 s.31 requires

		purge policy. Applied role-based access control (RBAC) restricting access to authorised roles only.	data be kept no longer than necessary. Indefinite retention compounds breach risk and regulatory exposure.
4	Step 5 — Preprocessing Service	Defined PII Handling Rules: Field validation (reject malformed inputs), format standardisation (normalise phone, income fields), completeness checks (flag >20% missing for manual review), and pseudonymisation of direct identifiers before ML input.	No defined handling rules = dirty ML input = unreliable credit scores. Pseudonymisation at this stage also limits the model's direct exposure to raw PII.
5	Step 7 — Decision Service	Transparency Logging + Human Review Trigger: Every automated decision now logs applicant ID, outcome, credit score, contributing features, and timestamp. Borderline scores ($\pm 10\%$ of threshold) routed to human loan officer.	Fully automated un-logged decisions violate s.22 (right to explanation). Logs are also the prerequisite for LARDR bias monitoring — without them no fairness audit is possible.
6	Steps 9 & 10 — Analytics DB & 3rd-Party Partner	PII Masking / Anonymisation: Applied irreversible anonymisation (aggregation / k-anonymity) before data enters Analytics DB or is shared externally. Direct identifiers removed or hashed. Third-party agreements updated to restrict re-identification.	Storing and sharing raw PII in analytics layers violates both data minimisation (s.17) and security principles. Any compromise of the Analytics DB or partner system would directly expose customer financial identity data.

Deliverable 3 — Summary of Review Process

My review of QuickLoan Mobile's data pipeline began by mapping the full data lifecycle — from collection at the User Mobile App through to storage and third-party sharing — against two analytical frameworks: the Data Lifecycle Model (Collect, Store, Process, Share, Archive, Destroy) and a four-tier Data Classification schema (Public, Internal, Confidential, Sensitive).

At the collection stage, applying the classification lens immediately flagged the pipeline's most critical flaw: the app was collecting Sensitive PII — financial records, contact data, and location — with no lawful basis and no consent mechanism, directly violating Act 843, Sections 14 and 17. Classification forced the question: "Does this data category need to exist here at all?" For GPS logs and full contact lists, the answer was clearly no, making data minimisation the first and most urgent correction.

Moving through the lifecycle, the absence of retention and purge policies at the Raw Data DB became visible through the Archive and Destroy phase of the lifecycle model. Sensitive data with no defined end-of-life accumulates indefinitely — a compounding liability that classification as Sensitive PII makes impossible to defer. The classification label is not just a label; it is a trigger for mandatory governance controls.

The bias risk emerged at the Process and Use stage. Once I identified that the Decision Service was operating with no logging and full automation, the structural absence of any feedback loop for detecting discriminatory outcomes became clear: you cannot correct what you cannot measure. The LoanScore model's use of proxy variables (neighbourhood, device type) encodes historical lending inequalities invisibly into every automated decision.

This is precisely why the Loan Approval Rate Disparity Ratio (LARDR) metric is the centrepiece of the governance recommendation. By computing the ratio of approval rates across demographic segments each month and visualising it as a Grouped Bar Chart, QuickLoan creates a standing accountability mechanism. A single number translates the ethical commitment to fairness into a monthly governance signal that board members, regulators, and auditors can understand and act on — without needing to interpret the ML model's internals. It closes the loop between automated decision-making and human oversight, ensuring that algorithmic bias cannot persist invisibly through the data lifecycle.